→JS-4

Loops

Used to iterate a piece of code

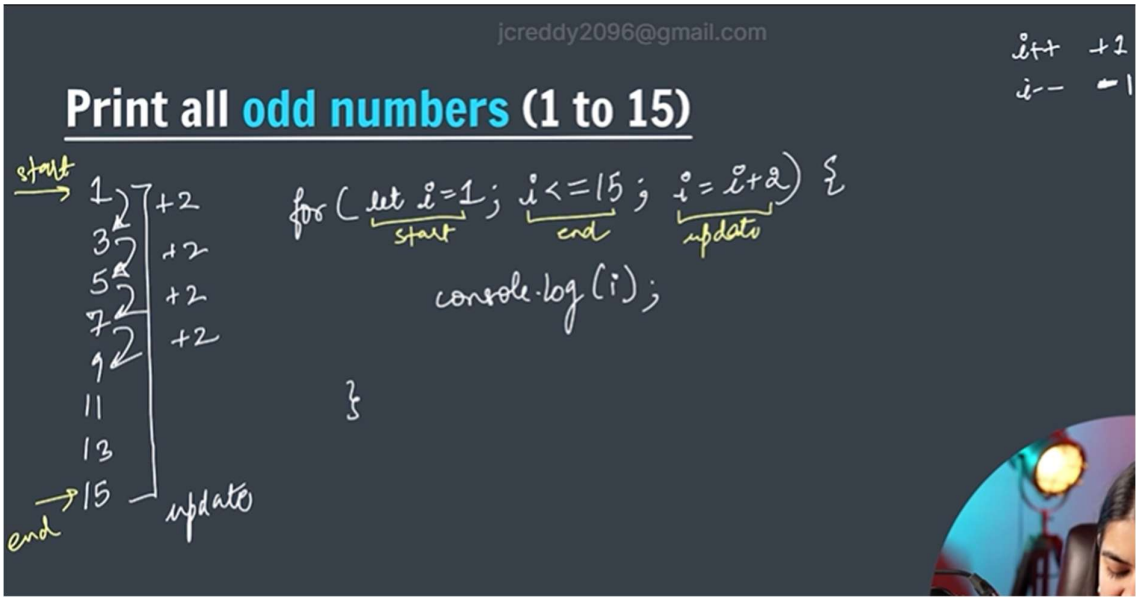
→For loops

for loop

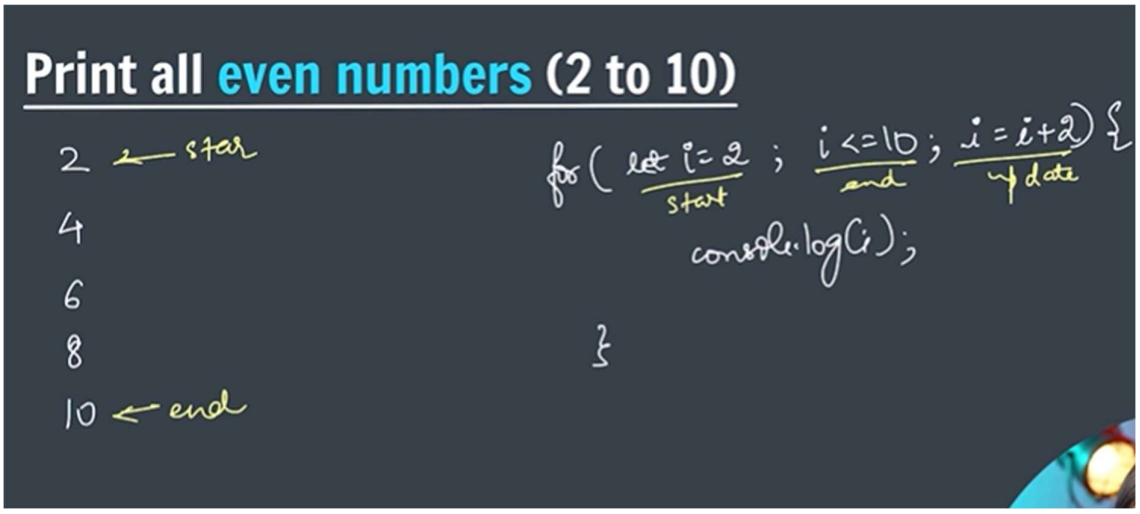
```
for(initialisation; condition; updation) {
    //do something
}


for (let i=1; i<=5; i++) {
    console.log(i);
}
```

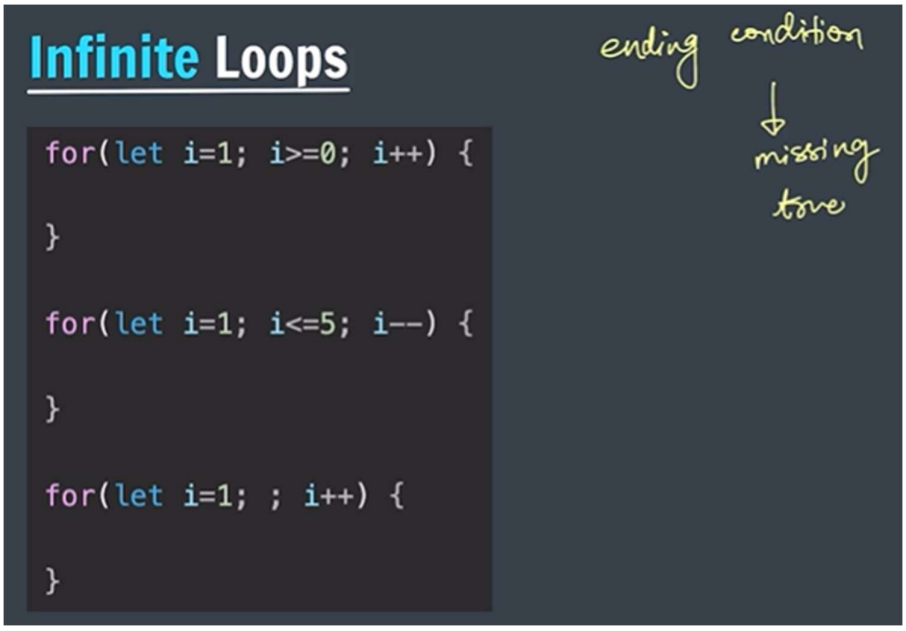
→Print all Odd numbers 1 to 15



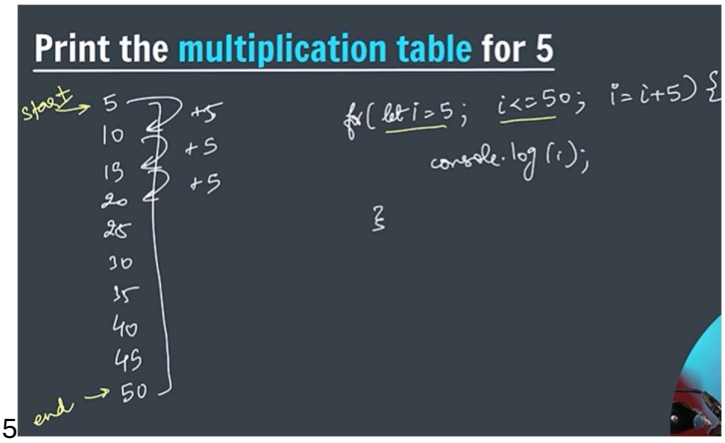
→Print even numbers



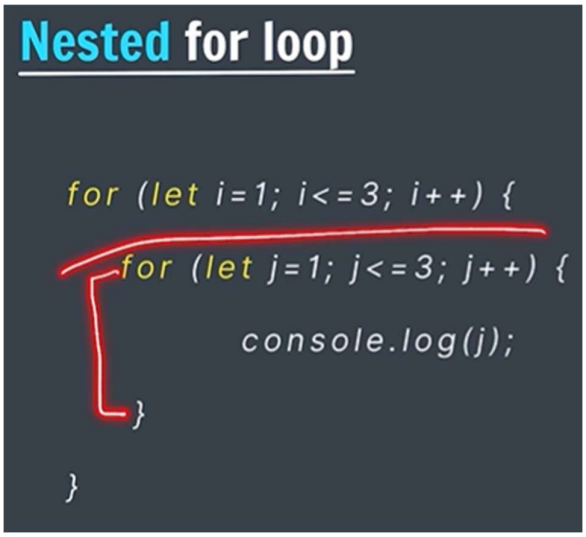
→Infinite loops



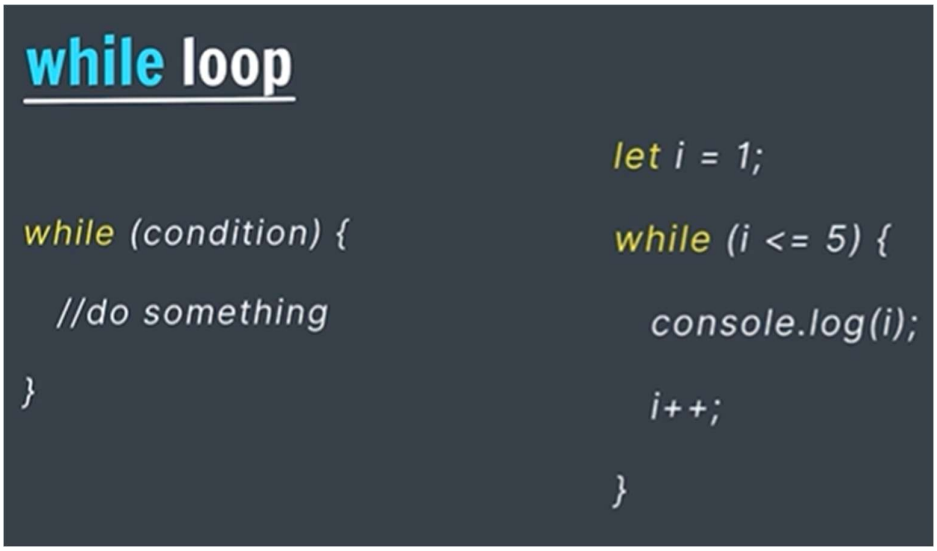
→print multiplication of 5

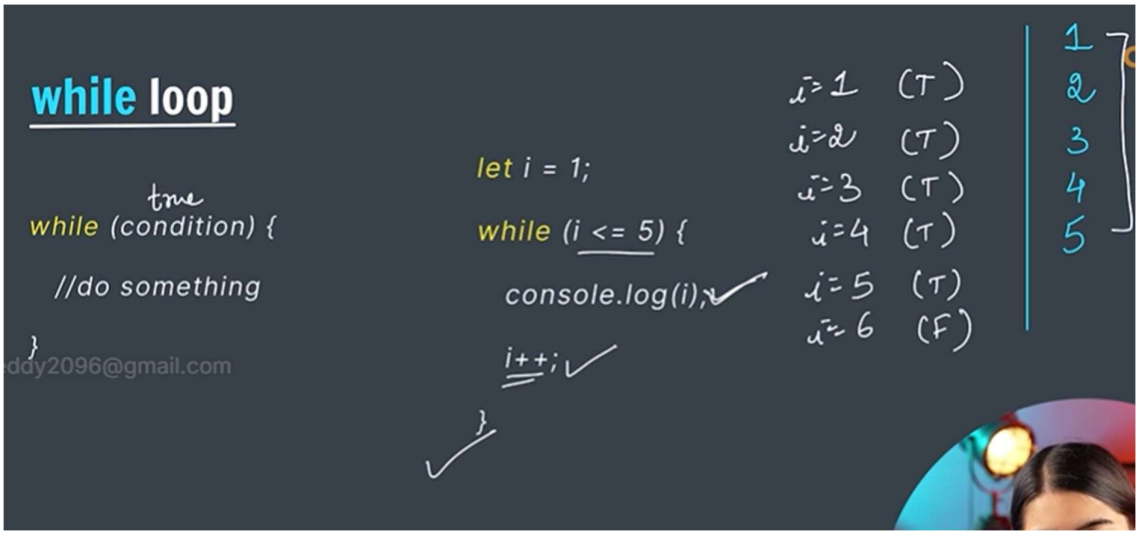


→nested for loop

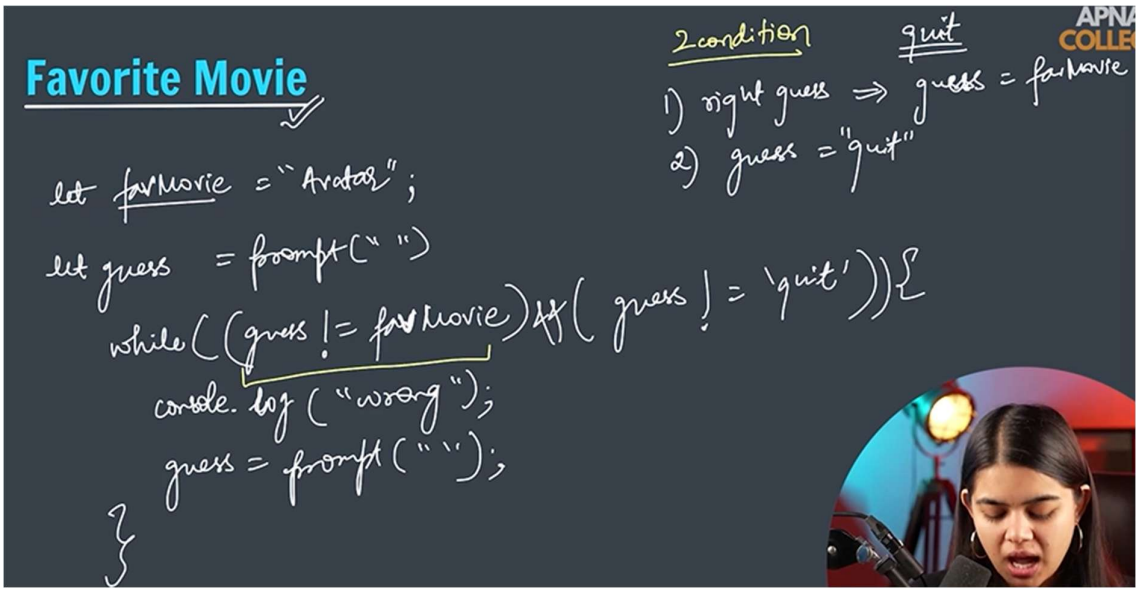


→While loop



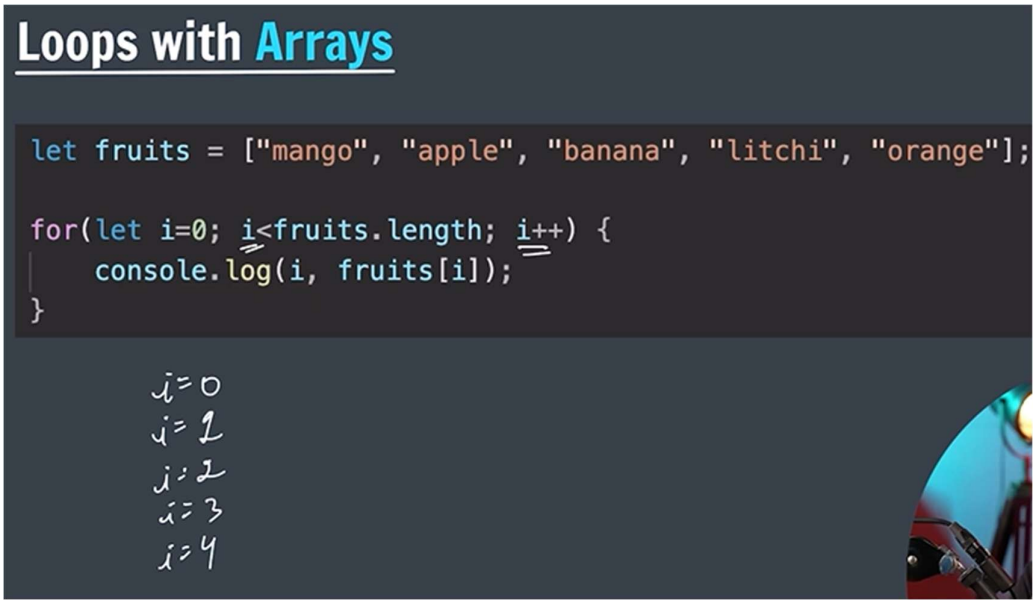


→Favorite movie

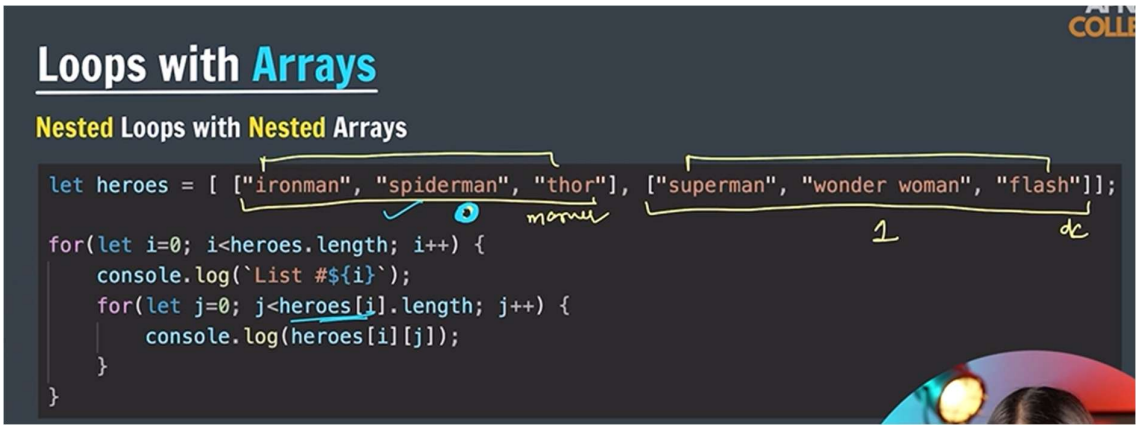


→break keyword

→Loops with Arrays

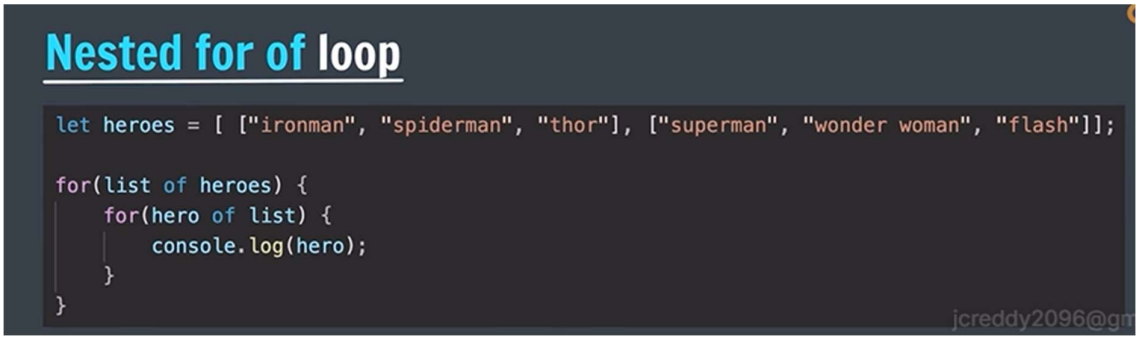


→Nested loop Arrays



→For of loop

→for of loops in nested loop



→JS-5

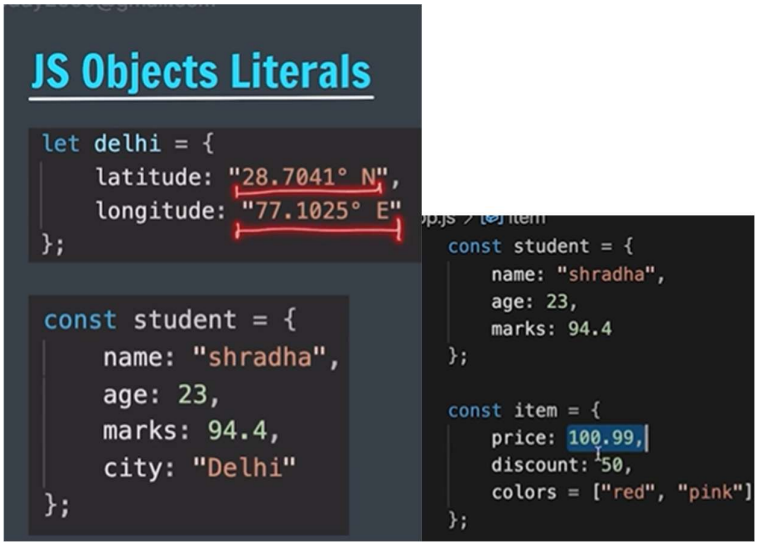
→Object Literals


```
let student = {
    name: "shradha",
    age: 23,
    marks: 94.4
};

let student2  = ["shradha", 23, 94.4];
```

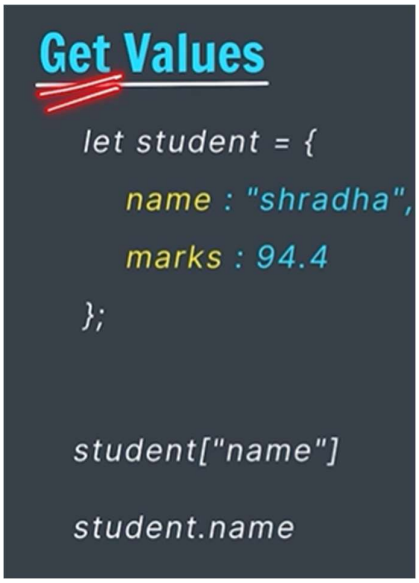
→we get the clarity in object literals than arrays.

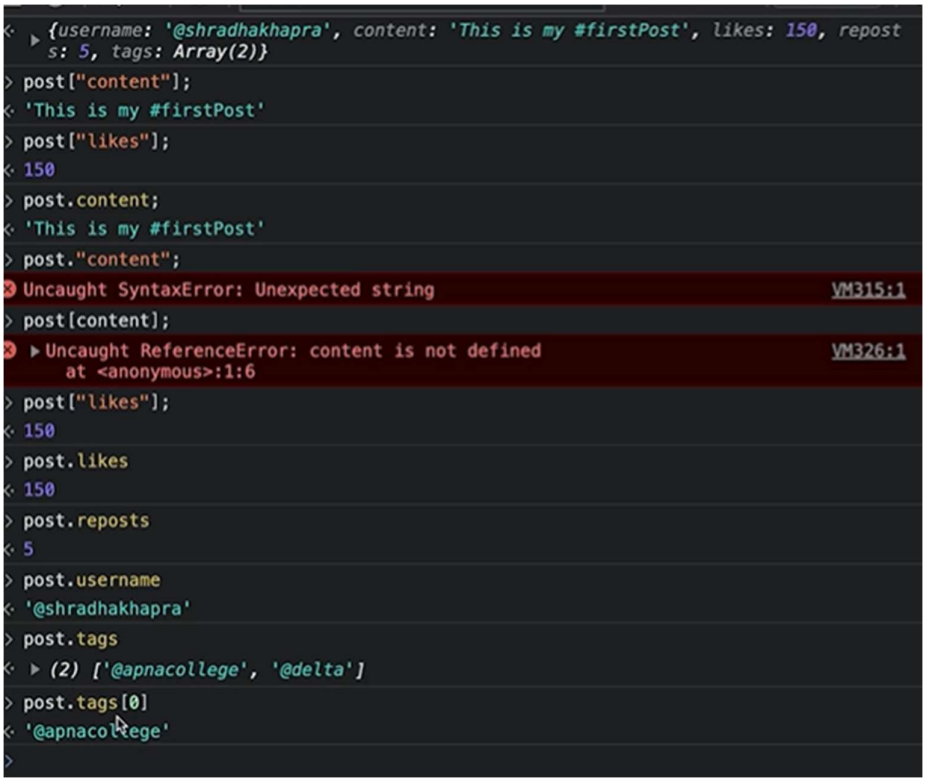
```
const student = {
    name: "shradha",
    age: 23,
    marks: 94.4
};
```



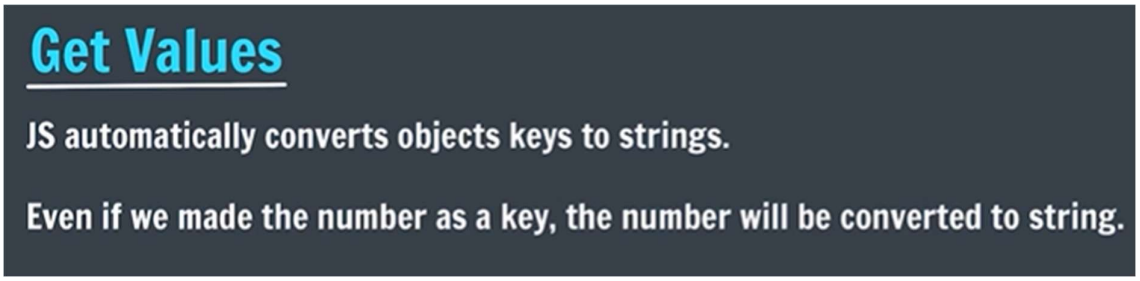
→Thread/Twitter post

→Get values




{username: '@shradhakhapra', content: 'This is my #firstPost', likes: 150, reposts: 5, tags: Array(2)}
post["content"];
'This is my #firstPost'
post["likes"];
150
post.content;
'This is my #firstPost'
post."content";
Uncaught SyntaxError: Unexpected string
VM315:1
post[content];
Uncaught ReferenceError: content is not defined
at <anonymous>:1:6
VM326:1
post["likes"];
150
post.likes
150
post.reposts
5
post.username
'@shradhakhapra'
post.tags
(2) ['@apnacollege', '@delta']
post.tags[0]
'@apnacollege'

→Get values


Get Values
JS automatically converts objects keys to strings.
Even if we made the number as a key, the number will be converted to string.

→App/update value

Add/ Update Value

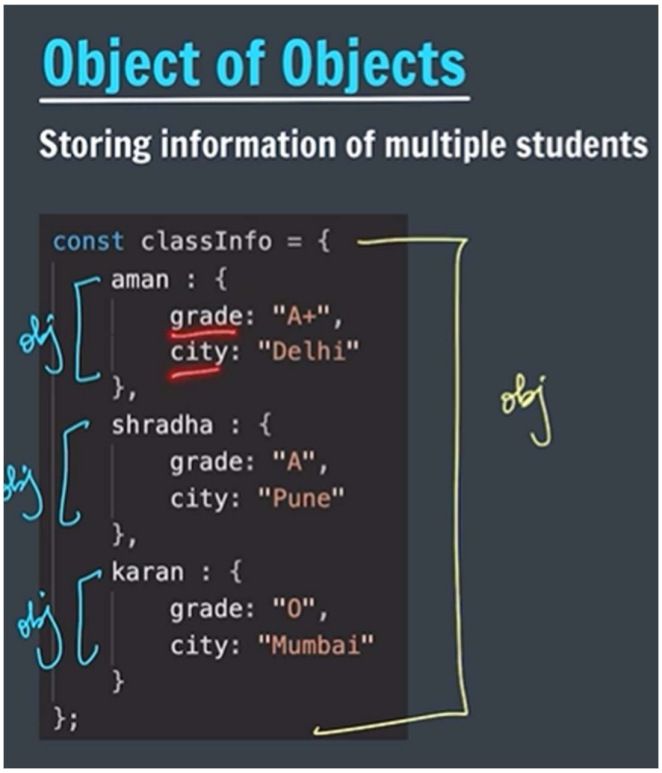
- Change the city to "Mumbai"
- Add a new property, gender: "Female"
- Change the marks to "A"

```
const student = {
    name: "shradha",
    age: 23,
    marks: 94.4,
    city: "Delhi"
};
```

```
student.gender = "female";
'female'
student.gender
'female'
student
▶ {name: 'shradha', age: 23, marks: 94.4, city: 'Mumbai', gender: 'female'}
student.marks
94.4
student.marks = "A";
'A'
student.marks
'A'
student
▶ {name: 'shradha', age: 23, marks: 'A', city: 'Mumbai', gender: 'female'}
student.marks = [99, 89, 74];
▶ (3) [99, 89, 74]
student
▼ {name: 'shradha', age: 23, marks: Array(3), city: 'Mumbai', gender: 'female'}
    age: 23
    city: "Mumbai"
    gender: "female"
  ▶ marks: (3) [99, 89, 74]
    name: "shradha"
  ▶ [[Prototype]]: Object
```

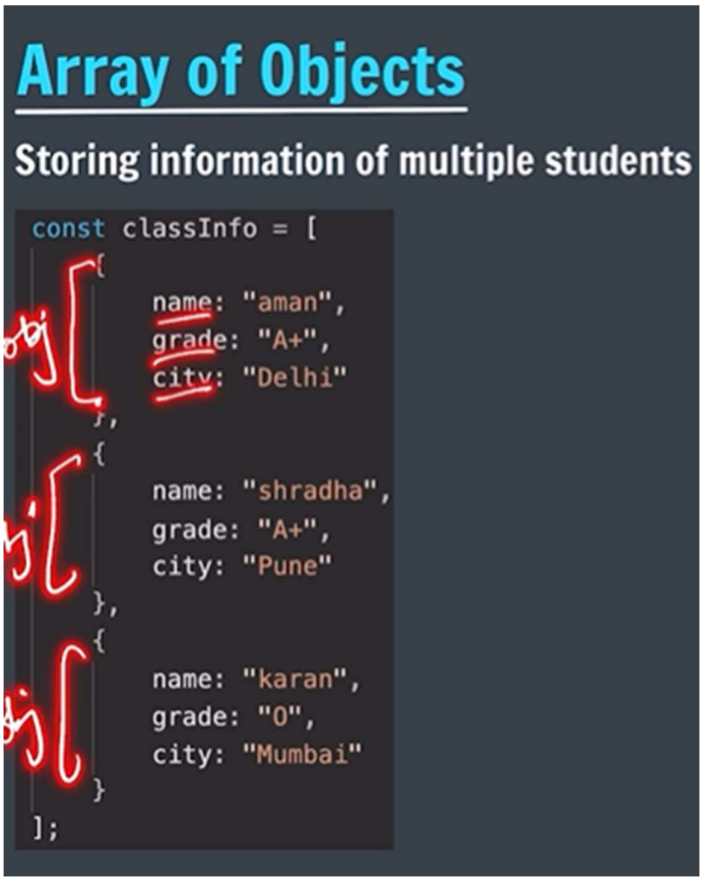
```
> student
< ▶ {name: 'shradha', age: 23, marks: 94.4, city: 'Delhi'}
> delete student.marks;
< true
> student
< ▶ {name: 'shradha', age: 23, city: 'Delhi'}
> delete student.city;
< true
> student
< ▶ {name: 'shradha', age: 23}
>
```

→Object of objects



```
> student
< ▶ {name: 'jai', age: 19, marks: 96, money: '1lakh'}
> classInfo
< ▼ {vinayaka: {…}, jai: {…}, siva: {…}} i
    ▶ jai: {grade: 'A', city: 'bkpalli'}
    ▶ siva: {grade: 'A', city: 'bkpalli'}
    ▶ vinayaka: {grade: 'A+', city: 'kanipakam'}
    ▶ [[Prototype]]: Object
> classInfo.vinayaka
< ▶ {grade: 'A+', city: 'kanipakam'}
> classInfo.siva.city
< 'bkpalli'
> classInfo.siva.city="rajyasabha"
< 'rajyasabha'
> classInfo
< ▼ {vinayaka: {…}, jai: {…}, siva: {…}} i
    ▶ jai: {grade: 'A', city: 'bkpalli'}
    ▶ siva: {grade: 'A', city: 'rajyasabha'}
    ▶ vinayaka: {grade: 'A+', city: 'kanipakam'}
    ▶ [[Prototype]]: Object
>
```

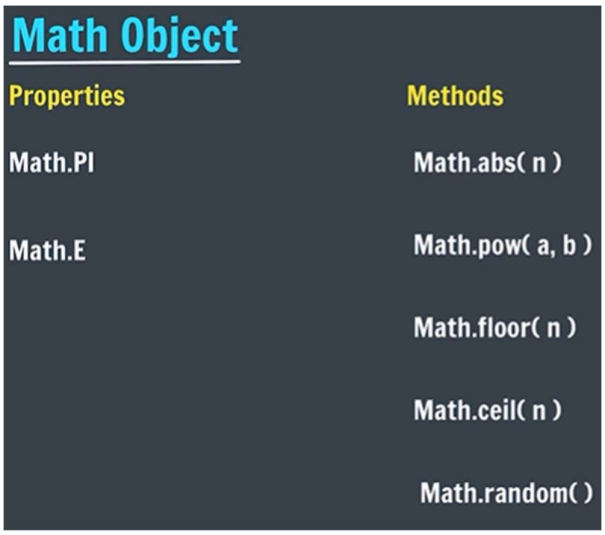
→Array of Objects



```
classInfo
▼ (3) [{…}, {…}, {…}]
  ▶ 0: {name: 'aman', grade: 'A+', city: 'Delhi'}
  ▶ 1: {name: 'shradha', grade: 'A', city: 'Pune'}
  ▶ 2: {name: 'karan', grade: 'O', city: 'Mumbai'}
    length: 3
  ▶ [[Prototype]]: Array(0)
classInfo[0]
▶ {name: 'aman', grade: 'A+', city: 'Delhi'}
classInfo[1]
▶ {name: 'shradha', grade: 'A', city: 'Pune'}
classInfo[2]
▶ {name: 'karan', grade: 'O', city: 'Mumbai'}
classInfo[1].name
'shradha'
classInfo[1].grade
'A'
classInfo[1].city
'Pune'
classInfo[1].city = "Gurgaon";
'Gurgaon'
classInfo[1]
▶ {name: 'shradha', grade: 'A', city: 'Gurgaon'}
```

```
classInfo[1]
▶ {name: 'shradha', grade: 'A', city: 'Gurgaon'}
classInfo[1].gender = "female";
'female'
classInfo[1]
▶ {name: 'shradha', grade: 'A', city: 'Gurgaon', gender: 'female'}
```

→Math objects



```
> Math
< ▶ Math {abs: ƒ, acos: ƒ, acosh: ƒ, asin: ƒ, asinh: ƒ, …}
> Math.PI
< 3.141592653589793
> Math.E
< 2.718281828459045
> Math.abs(12);
< 12
> Math.abs(-12);
< 12
> Math.abs(-12.5);
< 12.5
> Math.pow(2, 4);
< 16
> Math.pow(2, 5);
< 32
```

→Math.floor:it will give the numbers which is smaller and equal to the value :<= for floor

```
> Math.floor(5)
< 5
> Math.floor(5.5)
< 5
> Math.floor(5.9999999999)
< 5
> Math.floor(-5)
< -5
> Math.floor(-5.5)
< -6
>
```

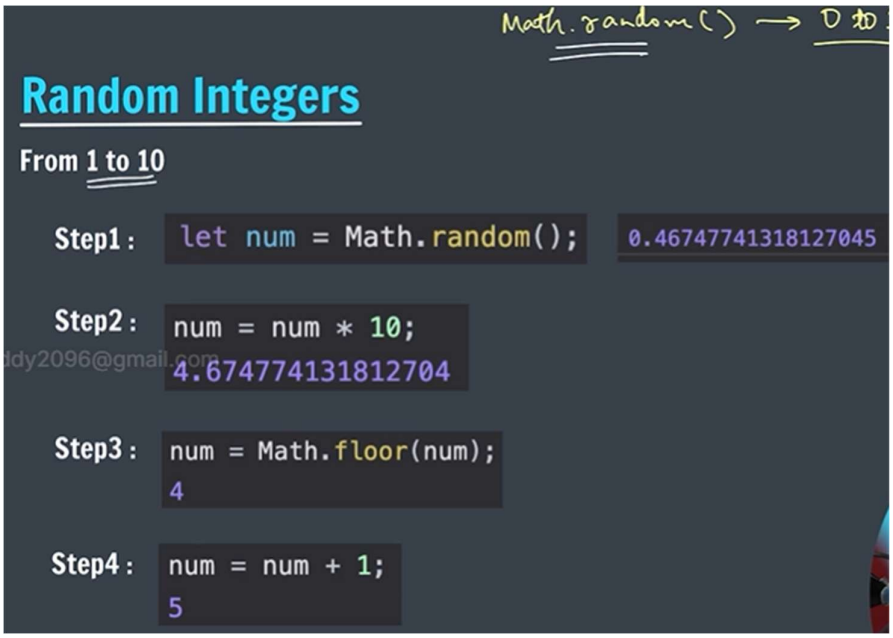
→Math.ceil : it gives the number greater than the given number

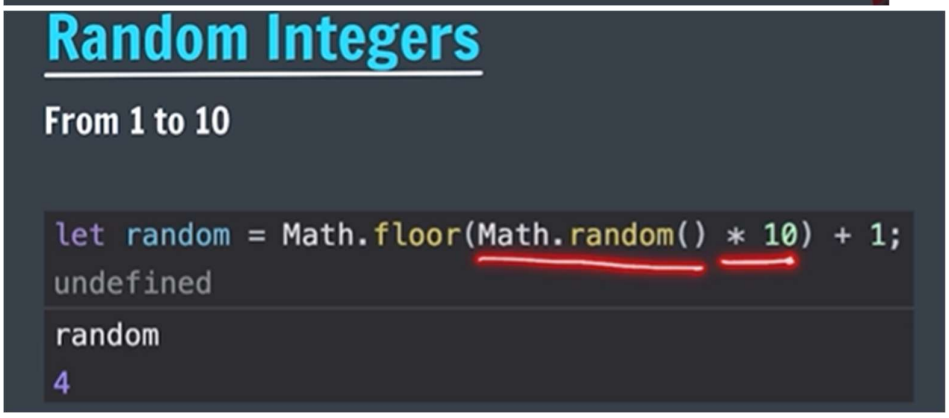
```
> Math.ceil(5)
< 5
> Math.ceil(5.5)
< 6
> Math.ceil(5.000000001)
< 6
> Math.ceil(-5)
< -5
> Math.ceil(-5.5)
< -5
```

→Math.random: it will give the value between the 0 to 1.

```
Math.random()
0.10191202663404164
Math.random()
0.13987166722029798
Math.random()
0.8144614591017076
Math.random()
0.7534498216923924
Math.random()
0.7413426254246709
Math.random()
0.6190270051920681
```

→Random Integers





→Practice Qs

Qs

max

Generate a random number between 1 and 100.

Math.floor(Math.random() * 100) + 1

0 to 99

5

Generate a random number between 1 and 5.

21 to 25

21, 22, 23, 24, 25